

PG-Drag: Diffusion-Prior and Gradient-Controlled Mesh Dragging

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Abstract

We present PG-Drag, a diffusion-prior-enhanced mesh dragging framework for balancing semantic plausibility, geometric quality, and handle accuracy. Built upon APAP, PG-Drag introduces early SDS guidance, handle-neighborhood gradient control, Handle Freeze, and a Sparse Shape Prior (SSP). Experiments demonstrate improved semantic consistency and geometric quality over representative mesh deformation baselines.

Keywords: Mesh deformation; diffusion priors; drag-based editing; score distillation sampling

1. Introduction

Drag-based mesh deformation provides intuitive control for interactive 3D editing. Classical geometric methods such as ARAP [1] preserve local geometric structures but lack high-level semantic guidance. APAP [2] combines Neural Jacobian Fields (NJF) [3] with Score Distillation Sampling (SDS) [4], introducing diffusion priors into mesh dragging through a two-stage optimization framework.

However, the decoupling between geometric deformation and diffusion refinement in APAP may produce implausible initial shapes that are difficult to recover in later optimization. Moreover, SDS gradients can cause local spikes and excessive stretching near manipulation handles, pull aligned handles away from their targets, and introduce unintended deformation in non-edited regions. To address these issues, PG-Drag introduces lightweight SDS guidance during the initial handle-alignment stage and spatially attenuates SDS gradients near the handles. After alignment, a Handle Freeze mechanism stabilizes the manipulated vertices, while a Sparse Shape Prior (SSP) constrains surface drift and spatial occupancy during semantic refinement. Together, these designs improve semantic plausibility while maintaining dragging accuracy and geometric stability.

2. Method

Given a source mesh $\mathcal{M}_0 = (V_0, F_0)$, handle vertices K_h , and targets T_h , PG-Drag follows the Jacobian-field deformation framework of APAP. Handle alignment is optimized by

$$\mathcal{L}_h = \|K_h V^* - T_h\|_2^2. \quad (1)$$

Stage I. Lightweight SDS is introduced during handle alignment to improve semantic plausibility. To avoid SDS-induced artifacts near manipulated handles, its gradients are spatially controlled:

$$(\nabla_V \mathcal{L}_{\text{SDS}})_{\text{masked}} = \mathbf{M} \odot \nabla_V \mathcal{L}_{\text{SDS}}. \quad (2)$$

Stage II. After alignment, Handle Freeze stabilizes the manipulated vertices, while SSP constrains global geometric drift:

$$\mathcal{L}_{\text{SSP}} = \alpha \mathcal{L}_{\text{surf}} + \beta \mathcal{L}_{\text{io}}. \quad (3)$$

The two-stage objectives are

$$\mathcal{L}_I = \lambda_h \mathcal{L}_h + \lambda_{\text{sds}}^{(1)} \mathcal{L}_{\text{SDS}}, \quad (4)$$

$$\mathcal{L}_{II} = \lambda_{\text{sds}}^{(2)} \mathcal{L}_{\text{SDS}} + \lambda_{\text{ssp}} \mathcal{L}_{\text{SSP}} + \mathcal{L}_{\text{freeze}}. \quad (5)$$

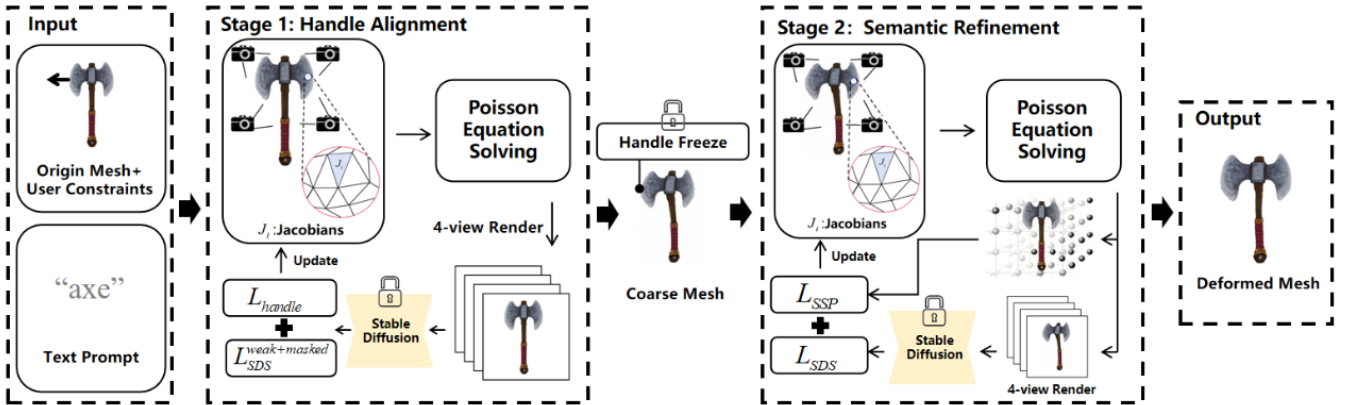


Figure 1: Overview of PG-Drag. Stage I performs gradient-controlled diffusion-assisted handle alignment, while Stage II combines Handle Freeze, SDS, and SSP for stable semantic refinement.

3. Experiments

We evaluate PG-Drag on ten mesh models from the APAP [2] repository and compare it with APAP, Smooth-ARAP

[5], and Local Deformation [6]. Semantic consistency is measured using CLIP [7], geometric distortion using P99 Symmetric Dirichlet Energy E_{SD} [8], dragging accuracy using NME, and runtime under the same hardware setting.



Figure 2: Visual comparison with APAP, Smooth-ARAP, and Local Deformation. Red arrows indicate dragging directions and green dots denote fixed anchors.

Table 1: Quantitative comparison. Best results are shown in bold.

Method	CLIP $\uparrow$	E_{SD} (P99) $\downarrow$	NME $\downarrow$	Time $\downarrow$
Local Deformation	0.2592	5.3920	5.51%	1 min
Smooth-ARAP	0.2564	4.5726	0.00%	1 min
APAP	0.2659	4.8534	3.87%	8 min
PG-Drag	0.2736	4.5545	1.12%	5 min

Quantitative results. PG-Drag achieves the highest CLIP score (0.2736) and the lowest E_{SD} (4.5545). It also reduces NME from 3.87% in APAP to 1.12%, demonstrating a better balance between semantic plausibility, geometric quality, and control accuracy.

4. Conclusion

PG-Drag improves diffusion-guided mesh dragging through early SDS guidance, gradient control, Handle Freeze, and SSP. Experiments demonstrate improved semantic plausibility, geometric quality, and dragging accuracy over APAP and representative geometric baselines.

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